

STATISTICAL ANALYSIS PLAN

VERSION 2 2024/03/20

EMPROTECT

« Embolization of the middle meningeal artery for the prevention of chronic subdural hematoma recurrence in high-risk patients »

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1 Brief summary

Standard of care for the management of symptomatic chronic subdural hematomas (CSDH) is neurosurgical burr-hole evacuation followed by drainage. Post-operative recurrence rates may be as high as 10 to 20 %. In particular, recurrence rate increases with antiplatelet and anticoagulant therapy. Middle meningeal artery (MMA) embolization has been proposed as a potential treatment to prevent CSDH recurrence. The aim of the EMPROTECT (Embolization of the Middle Meningeal Artery for the Prevention of Chronic Subdural Hematoma Recurrence in High Risk Patients) trial is to assess the efficacy of MMA embolization, performed within 7 days of CSDH burr-hole surgery, in reducing the risk of CSDH recurrence at 6 months as compared with standard medical treatment in patients at high risk of postoperative recurrence.

EMPROTECT is a multicenter open randomized controlled trial with blinded endpoint assessment by an adjudication committee. Two follow-up visits are planned: One month intermediate visit and 6 month final visit.

2 Objectives and endpoints

2.1 Main objective and primary endpoint

The primary objective of the trial is to assess the efficacy of MMA embolization in reducing the risk of postoperative recurrence of SDH at 6 months after burr-hole surgery as compared to standard medical treatment in patients at high risk of post-operative recurrence.

The primary outcome is the rate of chronic subdural hematoma recurrence 6 months after index burr-hole surgery, defined as:

- Reappearance of a homolateral CSDH with a midline shift > 5mm or a symptomatic homolateral SDH, including leading to death
- Or, the presence of a homolateral CSDH > 10mm in maximal thickness on the 6 months control head CT scan
- Or, the need for repeated surgery for a homolateral CSDH recurrence
- Or, the need for a new hospital admission in relation to a homolateral CSDH recurrence

A blinded assessment of all potential recurrences according to the definition of the primary endpoint will be ensured by an independent adjudication committee.

The committee will adjudicate the following cases:

- A recurrence following the definition reported in this protocol according to the local investigator
- A hospital admission potentially related to the CSDH
- A medical visit other than the M1 and M6 planned visits, potentially related to the CSDH

- A head CT or MRI scan, other than those planned at M1 and M6, potentially related to a CSDH recurrence
- Any new surgery potentially related to the CSDH
- Any neurological symptoms.

2.2 Secondary endpoints

The secondary endpoints are the following:

- Rate of repeated surgery for a homolateral CSDH recurrence during the 6-month follow-up period
- Rate of disability and dependency at 1 and 6 months, defined by a mRS score ≥ 4 . The mRS is an ordered scale coded from 0 (no symptoms at all) through 5 (severe disability) to 6 (death)
- Mortality rate at 1 and 6 months
- Total cumulative duration of hospital stay, during the 6-month follow-up period, directly or indirectly related to the CSDH
- Minor and major embolization procedure-related complication rates.
 - Major complications are defined as :
 - Major complications at the puncture site (retroperitoneal hematoma or arterial pseudo-aneurysm requiring surgical treatment)
 - Deterioration of renal function with a persistent drop in creatinine clearance of more than 10 points
 - Neurological accident with permanent sequelae
 - Other major complications
 - Minor complications are defined as:
 - Superficial hematoma at the puncture site
 - Transient deterioration of renal function
 - Transient neurological accident or asymptomatic arterial dissection
 - Other minor complications

3 Methods

3.1 Flow chart

A CONSORT flow chart will be provided to describe the number of eligible patients and the number of patients effectively included and randomized in total and in each of the two arms as well as patient follow-up (number lost to follow-up per arm). It will also present the number of patients analyzed in each group.

3.2 Sample size calculation

In order to show a reduction of the primary endpoint (6-month CSDH recurrence rate) from 15% (control group) to 5% (intervention group) with an 80% power, an overall bilateral alpha (α) risk

of 5%, 142 patients per group are necessary. The sample size has been calculated considering two sequential tests planned according to the Lan and Demets method with an O'Brien & Fleming α -spending function with an interim analysis after the end of the 6-month follow-up period of the first 129 patients representing a 37.5% information fraction, at the nominal α threshold of 0.001 and a final analysis at the end of the study at the α threshold of 0.05. Assuming a loss to follow-up rate of 20%, the number required is 342 patients in total (171 per group).

3.3 Analysis population

The primary analysis will be carried out in the intention-to-treat (ITT) population; each patient randomized in the study will be analyzed in the arm assigned by randomization regardless of the treatment actually received.

A sensitivity analysis will be carried out by excluding patients from the experimental group who did not receive embolization following the results of the CT angiography.

3.4 Descriptive analysis

For each group and at each evaluation time, the categorical variables will be described as number, percentage and missing data by response modality and the quantitative variables by mean, standard deviation or median and interquartile range (IQR) according to the normality distribution of the variable.

3.5 Primary endpoint analysis

The primary endpoint will be analyzed in 2 stages sequential design:

1. An interim analysis planned for 129 randomized patients with known recurrence status at 6 months.
2. A final analysis at the end of the study

This sequential analysis is based on a 2-step Lan and Demets approach with an O'Brien & Fleming type alpha risk spending function.

Interim analysis

For the interim analysis, 6-month CSDH recurrences by group will be analyzed using a mixed logistic model regression adjusted on randomization stratification factors. The randomization group (intervention versus control), the use of anticoagulant/antiplatelet and the unilateral or bilateral nature of the CSDH will be included as fixed effects and the center will be introduced as a random effect. In case of non-convergence of the model, a Mantel-Haensel test (adjusted on the center and other randomization stratification factors) will be used.

Patients deceased without prior recurrence will be considered as recurrences.

The primary analysis will be carried out according to the results of the adjudication and after imputation, which we will be completed with the analysis without adjudication of the recurrence of CSDH (assessment of the primary outcome by on-site investigators), in sensitivity analysis.

The conditional power will also be estimated by simulations with the assumptions leading to the calculation of the necessary number of subjects knowing the recurrence rates observed during the interim analysis. It will be given for information to the independent Data Safety Monitoring Board (DSMB). Stopping the trial for futility could possibly be decided if the conditional power is considered by the DSMB to be excessively low. However, no futility limit is provided.

In the event of a decision to stop the study, following the interim analysis, for rejection of the hypothesis of equality of CSDH recurrences rates between the two groups at the nominal threshold of alpha at 0.001, the final analysis will include the patients analyzed who led to the decision to stop, as well as all the patients for whom the primary endpoint could be collected subsequently (over-running patients). Likewise, we will continue to monitor included patients who have already benefited from embolization until 6 months and they will be included in the final analysis.

The interim analysis has been carried out and the results did not meet the predefined stopping rule. Therefore, the DSMB recommended to continue the trial.

Final analysis of the primary endpoint

- **Primary analysis**

The primary analysis will be the analysis of the primary outcome as assessed by the adjudication committee, in the intention-to-treat population, using a mixed logistic regression model adjusted on randomization stratification factors. The randomization group (intervention versus control), the use of anticoagulant/antiplatelet, the unilateral or bilateral nature of the SDH will be included as fixed effects and the center will be introduced as a random effect (to take into account the multicenter nature of the study). An adjusted absolute risk difference, in the recurrence rate at 6 months, will also be estimated as well as its 95% confidence interval using bootstrap.

Missing data will be imputed using multiple imputation and patients deceased without recurrence, with a cause of death considered to be neurological or undetermined, will be considered as treatment failure.

- **Subgroup analyses**

Two subgroup analyses for the primary outcome will be performed according to the unilateral versus bilateral CSDH status and anticoagulant/antiplatelet therapy medication status (yes vs no). For the two analyses, we will test interaction between randomization group and the two variables, respectively, in the logistic model describe above.

- **Sensitivity analyses**

The primary analysis will be completed with sensitivity analyses:

- Analysis without adjudication of the recurrence of CSDH (assessment of the primary outcome by on-site investigators)
- Analysis in the intent to treat population but without imputation (ie, lost to follow-up will be considered as no event)
- Analysis without patients who did not receive embolization in the experimental arm (following CT angiography or for other reasons) if any (ie, modified intent to treat analysis)
- Analysis without patients wrongly included (ie, patients presenting at least one non-inclusion criteria or not all inclusion criteria) if any

3.6 Secondary endpoints analyses

- Rate of repeated surgery for a homolateral CSDH recurrence during the 6-month follow-up period will be compared between the two treatment groups using Chi-square test or Fisher exact test as appropriate.
- Rate of disability and dependency, defined by an mRS score ≥ 4 , at 1 month and 6 months will be compared between the two treatment groups using a GEE logistic model. This model will include as parameters the randomization arm (intervention versus control), the time as categorical variable (1 and 6 months) and the time*randomization arm interaction. The interaction term will determine whether a significant difference is observed between the two arms of the study at one month and at 6 months. At each time (1 month and 6 months) and in each group, rate of disability and dependency will be estimated with its 95% confidence interval
- Mortality rate at 1 and 6 months will be compared between the two treatment groups using Chi-square test or Fisher exact test as appropriate
- Total cumulative duration of hospital stay, during the 6-month follow-up period, directly or indirectly related to the CSDH, will be compared between the two treatment groups using Wilcoxon rank-sum test.
- Minor and major complications will be compared between the two treatment groups using Chi-square test or Fisher exact test as appropriate.

Adverse events observed at 1 month and 6 months will be described overall and according to their nature by treatment group, in terms of number, and percentage of patients affected. Serious adverse events will be described separately. Fisher exact test will be used to compare the proportions of patients who died, with at least one adverse event/serious adverse event, at least one serious adverse event between the two treatment arms. The number of adverse events and the number of serious adverse events will be compared between the two groups using Poisson regression model.

3.7 Statistical significance level planned

All analyses will be carried out at the two-sided alpha risk of 5% (except for the interim analysis carried out at the nominal threshold of 0.001).

3.8 Statistical criteria for stopping research

Inclusions could be stopped after execution of the interim analysis, in the event of a difference between the two groups proven at the threshold of 0.001 or due to futility in the event of conditional power deemed excessively low.

The research will end at the end of the follow-up of the last patient included.

3.9 Missing data management

In the primary endpoint analysis, a multiple imputation method will be implemented for patients who cannot be evaluated for the main endpoint.

The Multiple Imputation by Chained Equations (MICE) technique will be used to replace missing values. The variables used in the imputation phase will be recurrence at 6 months, age, sex, unilateral or bilateral nature of CHDS and use of anticoagulant/antiplatelet as well as the characteristics of CSDH at inclusion, namely: Type of CSDH (initial episode or recurrence), proven recent head trauma, GCS score, ASA score, focal neurological deficit, hemiparesis, phasic disorder, gait disorder, headache, maximum thickness of the SDH, Park classification, type of surgery (burr hole, Trephine craniotomy), complications (infection, subdural empyema, epileptic seizure occurring during surgery or immediately afterward). Analyses will be carried out using R software and the mice package. 10 datasets will be generated. The results from each imputed dataset will be pooled using Rubin's rules to obtain overall estimates and their standard errors.

3.10 Different versions of the statistical analysis plan and changes made

Any modification to the initial analysis plan (SAP) will be subject to justification described in the study report.

- A first SAP version was carried out before database lock for the interim analysis
- This version is prepared before database lock for the final analysis

PAS Version	Date	Analysis	Modifications
1	2022/07/11	Interim	
2	2024/02/28	Final	1. Changing the definition of recurrence in case of death. Only death from neurological or undetermined cause will be considered as CSDH recurrence 2. Adding a sensitivity analysis: Analysis without patients wrongly included (ie, patients presenting at least one non-inclusion criteria or not all

			inclusion criteria) if any
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3.11 Software

All analyses will be performed using R software with the latest version.